設定管理與 IaC

Configuration Management & IaC

AWS Certified DevOps Engineer – Professional (DOP-C02) ｜ 官方考試權重 17%

第 1 題 （題庫 No.10 · 單選）

A DevOps engineer is creating an AWS CloudFormation template to deploy a web service. The web service will run on Amazon EC2 instances in a private subnet behind an Application Load Balancer (ALB). The DevOps engineer must ensure that the service can accept requests from clients that have IPv6 addresses. What should the DevOps engineer do with the CloudFormation template so that IPv6 clients can access the web service?

A. Add an IPv6 CIDR block to the VPC and the private subnet for the EC2 instances. Create route table entries for the IPv6 network, use EC2 instance types that support IPv6, and assign IPv6 addresses to each EC2 instance.

B. Assign each EC2 instance an IPv6 Elastic IP address. Create a target group, and add the EC2 instances as targets. Create a listener on port 443 of the ALB, and associate the target group with the ALB.

C. Replace the ALB with a Network Load Balancer (NLB). Add an IPv6 CIDR block to the VPC and subnets for the NLB, and assign the NLB an IPv6 Elastic IP address.

D. Add an IPv6 CIDR block to the VPC and subnets for the ALB. Create a listener on port 443. and specify the dualstack IP address type on the ALB. Create a target group, and add the EC2 instances as targets. Associate the target group with the ALB.

我的答案：_____

第 2 題 （題庫 No.47 · 單選）

A DevOps engineer used an AWS Cloud Formation custom resource to set up AD Connector. The AWS Lambda function ran and created AD Connector, but Cloud Formation is not transitioning from CREATE_IN_PROGRESS to CREATE_COMPLETE. Which action should the engineer take to resolve this issue?

A. Ensure the Lambda function code has exited successfully.

B. Ensure the Lambda function code returns a response to the pre-signed URL.

C. Ensure the Lambda function IAM role has cloudformation UpdateStack permissions for the stack ARN.

D. Ensure the Lambda function IAM role has ds ConnectDirectory permissions for the AWS account.

我的答案：_____

第 3 題 （題庫 No.97 · 單選）

A DevOps engineer updates an AWS CloudFormation stack to add a nested stack that includes several Amazon EC2 instances. When the DevOps engineer attempts to deploy the updated stack, the nested stack fails to deploy. What should the DevOps engineer do to determine the cause of the failure?

A. Use the CloudFormation detect root cause capability for the failed stack to analyze the failure and return the event that is the most likely cause for the failure.

B. Query failed stacks by specifying the root stack as the ParentId property. Examine the StackStatusReason property for all returned stacks to determine the reason the nested stack failed to deploy.

C. Activate AWS Systems Manager for the AWS account where the application runs. Use the AWS Systems Manager Automation AWSSupport-TroubleshootCFNCustomResource runbook to determine the reason the nested stack failed to deploy.

D. Configure the CloudFormation template to publish logs to Amazon CloudWatch. View the CloudFormation logs for the failed stack in the CloudWatch console to determine the reason the nested stack failed to deploy.

我的答案：_____

第 4 題 （題庫 No.105 · 複選（Select 2））

A company uses an organization in AWS Organizations to manage multiple AWS accounts The company needs an automated process across all AWS accounts to isolate any compromised Amazon EC2 instances when the instances receive a specific tag. Which combination of steps will meet these requirements? (Select TWO.)

A. Use AWS Cloud Formation StackSets to deploy the Cloud Formation stacks in all AWS accounts.

B. Create an SCP that has a Deny statement for the ec2: " action with a condition of " aws:RequestTag /isolation " : false.

C. Attach the SCP to the root of the organization.

D. Create an AWS Cloud Formation template that creates an EC2 instance rote that has no 1AM policies attached. Configure the template to have a security group that has an explicit Deny rule on all traffic. Use the Cloud Formation template to create an AWS Lambda function that attaches the 1AM role to instances. Configure the Lambda function to add a network ACL. Sot up an Amazon EventBridge rule to invoke the Lambda function when a specific tag is applied to a compromised EC2 instance.

E. Create an AWS Cloud Formation template that creates an EC2 instance role that has no 1AM policies attached. Configure the template to have a security group that has no inbound rules or outbound rules. Use the CloudFormation template to create an AWS Lambda function that attaches the 1AM role to instances. Configure the Lambda function to replace any existing security groups with the new security group. Set up an Amazon EventBridge rule to invoke the Lambda function when a specific tag is applied to a compromised EC2 instance.

我的答案：_____

第 5 題 （題庫 No.120 · 複選（Select 3））

A company deploys its corporate infrastructure on AWS across multiple AWS Regions and Availability Zones. The infrastructure is deployed on Amazon EC2 instances and connects with AWS loT Greengrass devices. The company deploys additional resources on on-premises servers that are located in the corporate headquarters. The company wants to reduce the overhead involved in maintaining and updating its resources. The company ' s DevOps team plans to use AWS Systems Manager to implement automated management and application of patches. The DevOps team confirms that Systems Manager is available in the Regions that the resources are deployed m Systems Manager also is available in a Region near the corporate headquarters. Which combination of steps must the DevOps team take to implement automated patch and configuration management across the company ' s EC2 instances loT devices and on-premises infrastructure? (Select THREE.)

A. Apply tags lo all the EC2 instances. AWS loT Greengrass devices, and on-premises servers. Use Systems Manager Session Manager to push patches to all the tagged devices.

B. Use Systems Manager Run Command to schedule patching for the EC2 instances AWS loT Greengrass devices and on-premises servers.

C. Use Systems Manager Patch Manager to schedule patching loT the EC2 instances AWS loT Greengrass devices and on-premises servers as a Systems Manager maintenance window task.

D. Configure Amazon EventBridge to monitor Systems Manager Patch Manager for updates to patch baselines. Associate Systems Manager Run Command with the event lo initiate a patch action for all EC2 instances AWS loT Greengrass devices and on-premises servers.

E. Create an IAM instance profile for Systems Manager Attach the instance profile to all the EC2 instances in the AWS account. For the AWS loT Greengrass devices and on-premises servers create an IAM service role for Systems Manager.

F. Generate a managed-instance activation Use the Activation Code and Activation ID to install Systems Manager Agent (SSM Agent) on each server in the on-premises environment Update the

AWS loT Greengrass IAM token exchange role Use the role to deploy SSM Agent on all the loT devices.

我的答案：_____

第 6 題 （題庫 No.177 · 單選）

A large enterprise is deploying a web application on AWS. The application runs on Amazon EC2 instances behind an Application Load Balancer. The instances run in an Auto Scaling group across multiple Availability Zones. The application stores data in an Amazon RDS for Oracle DB instance and Amazon DynamoDB. There are separate environments tor development testing and production. What is the MOST secure and flexible way to obtain password credentials during deployment?

A. Retrieve an access key from an AWS Systems Manager securestring parameter to access AWS services. Retrieve the database credentials from a Systems Manager SecureString parameter.

B. Launch the EC2 instances with an EC2 1AM role to access AWS services Retrieve the database credentials from AWS Secrets Manager.

C. Retrieve an access key from an AWS Systems Manager plaintext parameter to access AWS services. Retrieve the database credentials from a Systems Manager SecureString parameter.

D. Launch the EC2 instances with an EC2 1AM role to access AWS services Store the database passwords in an encrypted config file with the application artifacts.

我的答案：_____

第 7 題 （題庫 No.229 · 單選）

A company runs an application on one Amazon EC2 instance. Application metadata is stored in Amazon S3 and must be retrieved if the instance is restarted. The instance must restart or relaunch automatically if the instance becomes unresponsive. Which solution will meet these requirements?

A. Create an Amazon CloudWatch alarm for the StatusCheckFailed metric. Use the recover action to stop and start the instance. Use an S3 event notification to push the metadata to the instance when the instance is back up and running.

B. Configure AWS OpsWorks, and use the auto healing feature to stop and start the instance. Use a lifecycle event in OpsWorks to pull the metadata from Amazon S3 and update it on the instance.

C. Use EC2 Auto Recovery to automatically stop and start the instance in case of a failure. Use an S3 event notification to push the metadata to the instance when the instance is back up and running.

D. Use AWS CloudFormation to create an EC2 instance that includes the UserData property for the EC2 resource. Add a command in UserData to retrieve the application metadata from Amazon S3.

我的答案：_____

第 8 題 （題庫 No.236 · 複選（Select 2））

A company updated the AWS Cloud Formation template for a critical business application. The stack update process failed due to an error in the updated template and AWS CloudFormation automatically began the stack rollback process Later a DevOps engineer discovered that the application was still unavailable and that the stack was in the UPDATE_ROLLBACK_FAILED state. Which combination of actions should the DevOps engineer perform so that the stack rollback can complete successfully? (Select TWO.)

A. Attach the AWSC loud Formation FullAccess IAM policy to the AWS CtoudFormation role.

B. Automatically recover the stack resources by using AWS CloudFormation drift detection.

C. Issue a ContinueUpdateRollback command from the AWS CloudFormation console or the AWS CLI.

D. Manually adjust the resources to match the expectations of the stack.

E. Update the existing AWS CloudFormation stack by using the original template.

我的答案：_____

第 9 題 （題庫 No.297 · 單選）

A company has developed a static website hosted on an Amazon S3 bucket. The website is deployed using AWS CloudFormation. The CloudFormation template defines an S3 bucket and a custom resource that copies content into the bucket from a source location. The company has decided that it needs to move the website to a new location, so the existing CloudFormation stack must be deleted and re-created. However, CloudFormation reports that the stack could not be deleted cleanly. What is the MOST likely cause and how can the DevOps engineer mitigate this problem for this and future versions of the website?

A. Deletion has failed because the S3 bucket has an active website configuration. Modify the Cloud Formation template to remove the WebsiteConfiguration properly from the S3 bucket resource.

B. Deletion has failed because the S3 bucket is not empty. Modify the custom resource ' s AWS Lambda function code to recursively empty the bucket when RequestType is Delete.

C. Deletion has failed because the custom resource does not define a deletion policy. Add a DeletionPolicy property to the custom resource definition with a value of RemoveOnDeletion.

D. Deletion has failed because the S3 bucket is not empty. Modify the S3 bucket resource in the CloudFormation template to add a DeletionPolicy property with a value of Empty.

我的答案：_____

第 10 題 （題庫 No.367 · 單選）

A company uses AWS Systems Manager to manage a fleet of Amazon Linux EC2 instances that have SSM Agent installed. All EC2 instances are configured to use Instance Metadata Service Version 2 (IMDSv2) and are running in the same AWS account and AWS Region. Company policy requires developers to use only Amazon Linux. The company wants to ensure that all new EC2 instances are automatically managed by Systems Manager after creation. Which solution will meet these requirements with the MOST operational efficiency?

A. Create an IAM role that has a trust policy that allows Systems Manager to assume the role. Attach the AmazonSSMManagedEC2InstanceDefaultPolicy policy to the role. Configure the default-ec2-instance- management-role SSM service setting to use the role.

B. Ensure that AWS Config is set up. Create an AWS Config rule that validates if an EC2 instance has SSM Agent installed. Configure the rule to run on EC2 configuration changes. Configure automatic remediation for the rule to run the AWS-InstallSSMAgent SSM document to install SSM Agent.

C. Configure Systems Manager Patch Manager. Create a patch baseline that automatically installs SSM Agent on all new EC2 instances. Create a patch group for all EC2 instances. Attach the patch baseline to the patch group. Create a maintenance window and maintenance window task to start installing SSM Agent daily.

D. Create an EC2 instance role that has a trust policy that allows Amazon EC2 to assume the role. Attach the AmazonSSMManagedInstanceCore policy to the role. Ensure that AWS Config is set up. Use the ec2-instance-profile-attached managed AWS Config rule to validate if an EC2 instance has the role attached. Configure the rule to run on EC2 configuration changes. Configure automatic remediation for the rule to run the AWS-SetupManagedRoleOnEc2Instance SSM document to attach the role to the EC2 instance.

我的答案：_____

解答與解析　Answer Key

解析取自題庫原文，未經 AWS 官方審核；若與官方文件有出入，以官方文件為準。

第 1 題（No.10）　正確答案：D

it involves adding an IPv6 CIDR block to the VPC and subnets for the ALB and specifying the dualstack IP address type on the ALB listener. This allows the ALB to listen on both IPv4 and IPv6 addresses, and forward requests to the EC2 instances that are added as targets to the target group associated with the ALB.

涉及服務：CloudFormation、EC2/Auto Scaling、ELB/ALB/NLB、VPC

第 2 題（No.47）　正確答案：B

Reference: https://docs.aws.amazon.com/AWSCloudFormation/latest/UserGuide/crpg-ref-responses.html

涉及服務：CloudFormation、Lambda、IAM/SCP

第 3 題（No.97）　正確答案：B

（此題題庫未附解析）

涉及服務：CloudFormation、Systems Manager (SSM)、EC2/Auto Scaling、CloudWatch

第 4 題（No.105）　正確答案：A、E

Step 1: Deploy the Automation Solution using CloudFormation StackSets
To automate the process across multiple AWS accounts within an organization, you can use AWS CloudFormation StackSets. StackSets allow you to deploy CloudFormation templates to multiple accounts within an organization, ensuring consistent infrastructure and automation.
Action: Use AWS CloudFormation StackSets to deploy the necessary resources across all AWS accounts.
This includes deploying the Lambda function and security groups that will isolate compromised EC2 instances.
Why: StackSets make it easy to deploy and manage resources across multiple AWS accounts, reducing the operational overhead.
Reference: AWS documentation on CloudFormation StackSets.
This corresponds to Option A: Use AWS CloudFormation StackSets to deploy the CloudFormation stacks in all AWS accounts.
Step 2: Isolate EC2 Instances using Lambda and Security GroupsWhen an EC2 instance is compromised, it needs to be isolated from the network. This can be done by creating a security group with no inbound or outbound rules and attaching it to the instance. A Lambda function can handle this process and can be triggered automatically by an Amazon EventBridge rule when a specific tag (e.g., " isolation ") is applied to the compromised instance.
Action: Create a Lambda function that attaches an isolated security group (with no inbound or outbound rules) to the compromised EC2 instances. Set up an EventBridge rule to trigger the Lambda function when the " isolation " tag is applied to the instance.
Why: This automates the isolation process, ensuring that any compromised instances are immediately cut off from the network, reducing the potential damage from the compromise.
Reference: AWS documentation on Tag-based Event Handling.
This corresponds to Option E: Create an AWS CloudFormation template that creates an EC2 instance role that has no IAM policies attached. Configure the template to have a security group that has no inbound rules or outbound rules. Use the CloudFormation template to create an AWS Lambda function that attaches the IAM role to instances. Configure the Lambda function to replace any existing security groups with the new security group. Set up an Amazon EventBridge rule to invoke

the Lambda function when a specific tag is applied to a compromised EC2 instance.

涉及服務：CloudFormation、Lambda、EC2/Auto Scaling、Organizations/Control Tower、IAM/SCP、EventBridge

第 5 題（No.120） 正確答案：**C、E、F**

https://aws.amazon.com/blogs/mt/how-to-centrally-manage-aws-iot-greengrass-devices-using-aws-systems-manager/?force_isolation=true

涉及服務：Systems Manager (SSM)、EC2/Auto Scaling、IAM/SCP、EventBridge

第 6 題（No.177） 正確答案：**B**

AWS Secrets Manager is a secrets management service that helps you protect access to your applications, services, and IT resources. This service enables you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. Using Secrets Manager, you can secure and manage secrets used to access resources in the AWS Cloud, on third-party services, and on-premises. SSM parameter store and AWS Secret manager are both a secure option. However, Secrets manager is more flexible and has more options like password generation.
Reference: https://www.1strategy.com/blog/2019/02/28/aws-parameter-store-vs-aws-secrets-manager/

涉及服務：Systems Manager (SSM)、EC2/Auto Scaling、Secrets Manager、RDS/Aurora、DynamoDB、ELB/ALB/NLB

第 7 題（No.229） 正確答案：**B**

https://aws.amazon.com/blogs/mt/how-to-set-up-aws-opsworks-stacks-auto-healing-notifications-in-amazon-cloudwatch-events/

涉及服務：CloudFormation、EC2/Auto Scaling、CloudWatch、S3、OpsWorks

第 8 題（No.236） 正確答案：**C、D**

https://docs.aws.amazon.com/cli/latest/reference/cloudformation/continue-update-rollback.html
For a specified stack that is in the UPDATE_ROLLBACK_FAILED state, continues rolling it back to the UPDATE_ROLLBACK_COMPLETE state. Depending on the cause of the failure, you can manually fix the error and continue the rollback. By continuing the rollback, you can return your stack to a working state (the UPDATE_ROLLBACK_COMPLETE state), and then try to update the stack again.

涉及服務：CloudFormation、IAM/SCP

第 9 題（No.297） 正確答案：**B**

Step 1: Understanding the Deletion FailureThe most likely reason why the CloudFormation stack failed to delete is that the S3 bucket was not empty. AWS CloudFormation cannot delete an S3 bucket that contains objects, so if the website files are still in the bucket, the deletion will fail.
Issue: The S3 bucket is not empty during deletion, preventing the stack from being deleted.
Step 2: Modifying the Custom Resource to Handle DeletionTo mitigate this issue, you can modify the Lambda function associated with the custom resource to automatically empty the S3 bucket when the stack is being deleted. By adding logic to handle the RequestType: Delete event, the function can recursively delete all objects in the bucket before allowing the stack to be deleted.
Action: Modify the Lambda function to recursively delete the objects in the S3 bucket when RequestType is set to Delete.
Why: This ensures that the S3 bucket is empty before CloudFormation tries to delete it, preventing the stack deletion failure.
Reference: AWS documentation on CloudFormation custom resources.
This corresponds to Option B: Deletion has failed because the S3 bucket is not empty. Modify the custom resource ' s AWS Lambda function code to recursively empty the bucket when RequestType is Delete.

涉及服務：CloudFormation、Lambda、S3

第 10 題（No.367） 正確答案：A

The most operationally efficient answer is to use Systems Manager Default Host Management Configuration.

This feature is designed specifically to make EC2 instances become Systems Manager managed instances automatically without manually attaching an IAM instance profile to every instance. AWS states that Default Host Management Configuration manages EC2 instances automatically as managed instances and uses a default IAM role for the account and Region. It also requires IMDSv2, which the scenario already satisfies.

Option A correctly creates the Systems Manager trusted role, attaches AmazonSSMManagedEC2InstanceDefaultPolicy, and configures the SSM service setting. Options B and D add AWS Config remediation complexity, and C uses Patch Manager incorrectly because SSM Agent is already installed and the requirement is automatic Systems Manager management.

涉及服務：Systems Manager (SSM)、EC2/Auto Scaling、Config、IAM/SCP